# Adaptive Cleft Lip Reconstruction via Deep Learning-Driven Style Transfer and Image Inpainting

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**Abstract**—Cleft lip is a congenital condition identified by a split in the upper lip, often associated with a cleft palate. Surgical interventions are usually done to achieve a functional and aesthetic restoration. Being able to foresee surgical results remains difficult, often requiring multiple treatments. In this project, we aim to predict post-surgical facial reconstruction outcomes for cleft lip patients through a generative image inpainting model. We create a synthetic infant dataset by applying style transfer on the CelebA-HQ dataset. This was then used to train the PUT transformer model for our facial image inpainting task. The outcome was a dataset of 30,000 synthetic infant images with an FID score of 70.24 against a children's dataset. These images did not directly represent any real infants, therefore bypassing ethical and privacy concerns. The image inpainting model achieves an LPIPS score of 0.1057 on cleft lip reconstruction tasks, producing visually plausible post-surgical predictions. Together, these contributions show that synthetic data generation combined with domain-adapted inpainting offers a usable, ethically motivated approach to cleft lip surgical outcome prediction.

**Index Terms**—Computer Vision, Deep Learning, Generative Model, Image Processing

## 1 Introduction

Orofacial clefts are among the most common congenital abnormalities [1], [2]. The most prevalent form is cleft lip, which is characterised by a gap or split in the upper lip [1], [3]. The condition ranges in severity from a small opening in the lip to a complete separation that extends from the lip to the nose, as seen in Figure 1. Cleft lip, as a congenital condition, develops while an infant is within the womb. It occurs when the fusion of numerous facial processes in the early embryonic development fails [1], [3]. This originates from both genetic and environmental factors [1], [4]. Many candidate genes and genetic loci have shown to be associated with orofacial clefts [4]. Additionally, environmental factors such as smoking, drinking alcohol, obesity and taking certain medications during pregnancy have also been shown to be factors in increasing the likelihood of orofacial clefts [1], [3], [4]. As such, the condition is entirely outside the control of the affected infant. With an estimated 183,000 new cases each year, cleft lip, often accompanied by cleft palate, affects approximately 1 in 700 live births worldwide [1], [5]. As of 2021, the total number of people living with it is estimated at 4.1 million [5]. The condition is approximately twice as common in males and shows significant variation across ethnic groups, with the highest rates observed in Asian and Native American populations and lowest in populations of African descent [1], [6]. The majority of infants with cleft lip are treated through surgery while they are 3 to 6 months old [3]. Surgical outcomes are generally favourable, with studies reporting acceptable results in over 90% of cases [7]. The majority of children who receive treatment go on to lead completely normal lives [3].

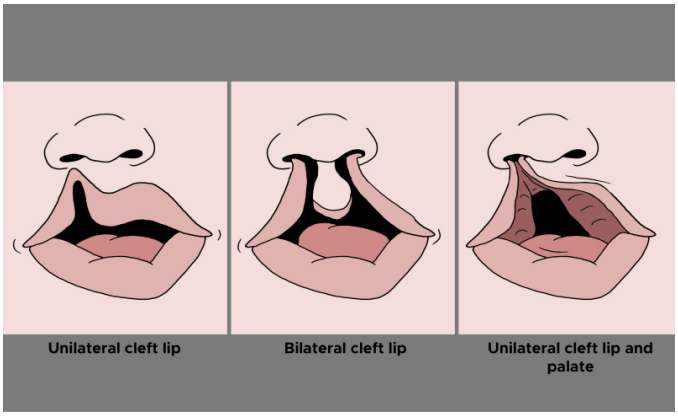



Fig. 1: Clinical presentation of cleft types: (a) unilateral cleft lip, (b) bilateral cleft lip, and (c) unilateral cleft lip and palate. Reproduced from Walker et al. [8].

However, many patients still require multiple procedures throughout childhood, with an average of 8 to 10 surgeries from birth to adulthood for those with cleft lip and palate [9].

Despite the high success rate of cleft lip surgery, predicting and visualising post-surgical outcomes remains a significant challenge for clinicians. Since surgery is a treatment with potential follow-up procedures, improving preparation and, therefore, the quality of the initial surgery should help reduce the need for follow-ups. A key component of treatment is the surgeons themselves, so being able to

support their preparation and give them another resource to help them physically visualise the surgical outcomes could lead them to improve their plans accordingly, identify potential issues in advance, and potentially reduce the need for revision surgeries. Despite the surgery's functional focus, cleft lip repair has a visible, aesthetic outcome that directly impacts the patient's face. Therefore, the outcome of the surgery is important as it can directly impact the patient's future self-esteem, social interactions and psychological wellbeing for the rest of their life.

Projects that try to address the challenge of predicting outcomes of cleft lip surgery face two key issues. First, there is no purely infant image dataset. Even when combining multiple facial datasets, there are still few infant images. This leaves researchers with the second issue of having insufficient data to train an infant image inpainting model using purely publicly available data. As a result, it is difficult to find an open-source infant-specific facial image inpainting model trained on publicly available data. One major barrier in the field of image inpainting is the lack of a publicly available infant-faces dataset due to ethical and privacy concerns. For a dataset based on real infant images, ethical issues stem from various aspects, such as data protection and privacy regulations designed to protect the infant's identity. In a facial recognition dataset, faces can be gradually recognised even under manipulations such as partial occlusion [10], meaning traditional anonymisation techniques like blurring are often not enough to hide identity. The collection, storage and sharing of images of minors for research are under strict control through global protection laws. Additionally, infants are unable to give consent for the collection of these images due to their limited cognitive abilities. This leads to their parents or guardians being responsible for giving consent for the infant's facial data to be stored. However, despite consent and methods to remove the infant's identity from the images through image editing, there is still the issue that the data remains re-identifiable, particularly for children with rare conditions, resulting in a privacy infringement and possible harm [10]. Beyond the dataset limitation, existing facial image inpainting models pose an additional challenge for this task. Current facial inpainting models are trained on datasets like CelebA-HQ and FFHQ. These datasets consist primarily of adult faces. Infant faces have different proportions compared to older age groups, for example, a larger forehead relative to face size, a smaller and rounder nose, different eye-to-face ratios and softer skin textures. A model trained on adult faces or a small number of infant faces would not be able to fully capture the facial characteristics of infants aged 2 or under. This discrepancy between the training data and the target domain introduces a significant domain shift, which is known to degrade model performance when applied to underrepresented groups or unseen data distributions [11]. Predicting the transformation of a cleft lip to a corrected lip is not a standard image inpainting task in which random missing regions are filled with plausible content. The model must specifically predict what the cleft lip region would look like after surgical repair, requiring an understanding of healthy infant facial structure in that area. This project addresses both of these challenges through two key contributions. Firstly, we create a synthetic infant facial dataset to sidestep ethical concerns entirely, as the images do not correspond to any real infants. This provides a complete infant dataset for future researchers facing the same data scarcity problem. Furthermore, we adapt and retrain an image inpainting model specifically on our synthetic infant data, allowing it to capture infant facial characteristics and perform the specialised task of predicting surgical outcomes.

To overcome these ethical and privacy issues, we developed a synthetic infant facial dataset. This was done via applying neural style transfer using the SAM (Only a matter of Style) model [12]. We used their StyleGAN-based architecture to perform age transformation on the faces from the CelebA-HQ dataset. This resulted in our dataset being 30,000 images with a resolution of 256x256, which do not correspond to any real individuals. We also retrained the PUT transformer-based image inpainting architecture [13] on our synthetic dataset to generate predicted post-surgical facial outcomes for cleft lip patients. While this infant-specific model works on general inpainting tasks on infant faces, we focus on its capabilities on inpainting a corrected cleft lip.

While existing research has explored facial image inpainting for general use, and in some cases using small privately available datasets, whether these models are able to predict cleft lip surgical outcomes using synthetic data remains unanswered. Therefore, this paper asks the following research question: Can a generative image inpainting model trained on synthetic infant data predict realistic post-surgical outcomes for cleft lip patients?

The remainder of this paper will address the related works for cleft lip post-surgical outcome prediction using inpainting models, as well as works regarding the generation of synthetic data in Section 2. Section 3 describes our proposed method to answer the research question. The results we obtained and our evaluations on them are in Sections 4 and 5, respectively. Section 6 concludes the paper. In summary, the main contributions of this paper are as follows:

- We present the first publicly available synthetic infant facial dataset (to our knowledge), containing 30,000 images at 256x256 resolution, generated using an iterative style-based age-regression pipeline on CelebA-HQ.
- We train the PUT transformer-based inpainting architecture on our synthetic infant dataset, adapting it from its original adult domain to the infant facial domain, enabling it to capture infant-specific facial characteristics.
- We demonstrate that our domain-adapted model can predict plausible post-surgical facial appearances for cleft lip patients, generalising from synthetic training data to real clinical images.

## 2 Related Work

### 2.1 Cleft lip visualisation and Surgical outcome prediction

The most directly relevant prior work to this project is the feasibility study by Chen et al., which explored whether

image inpainting could be used to generate a plausible non-cleft lip appearance from images of patients with cleft lip [14]. Chen et al. proposed that this method could address the challenge of predicting post-surgical outcomes by partially covering a patient's lip with a mask and performing inpainting, rather than relying solely on biomedical data. This shows that the cleft-lip outcome prediction task can be framed as a generative facial reconstruction problem, rather than a purely hypothetical application. However, Chen et al.'s approach was not trained on an infant-specific public dataset but on CelebA, a general facial dataset. Importantly, their results were evaluated on two small private clinical datasets totalling 34 real patient images from the Royal Victoria Infirmary (RVI) in Newcastle upon Tyne. This limits how far the results can be generalised, especially because infant faces differ from general face datasets in proportion, texture, and feature geometry [11].

Following Chen et al.'s feasibility study, later work extends this line of research in different directions. Atputharuban et al. [15] introduced CleftLipGAN, a structural-guided pluralistic inpainting model that included real-time interactive interface for adjusting outcomes. Like Chen et al., this work addresses post-operative cleft-lip reconstruction, but extends the task by having multiple plausible results be possible. However, the authors do highlight the recurring limitation of a scarcity of publicly available cleft-specific datasets. As a result, such models are typically trained on healthy facial images, such as FFHQ, and then applied to cleft conditions, which can lead to overly smoothed or structurally implausible outputs. A different but related direction is seen in the work of Rosero et al. [16], who focused on facial landmark detection for patients with repaired cleft lip and palate (CLP) rather than image generation. Rosero et al. argues that off-the-shelf landmark detectors trained on non-cleft faces fail on repaired CLP because they smooth out the lip shape but consequently miss important structural details. The authors propose CLP-Trans, a domain-transfer strategy that modifies public facial images to resemble repaired CLP outcomes while preserving privacy and reproducibility. Their best configuration reduces the normalised mean square error (NMSE) from 2.417 to 2.086 on patient images [16]. When combined, these studies show that cleft-lip-specific AI research is expanding across both reconstruction and facial analysis, but both directions still depend on synthetic or transferred representations because cleft-specific data remains limited.

### 2.2 Facial Image Inpainting for Structured Facial Reconstruction

At its core, image inpainting aims to restore corrupted or missing regions of an image by synthesising content that is coherent with the known surrounding pixels. While image inpainting can be applied to natural scenes and other visual contexts, its application to facial reconstruction introduces a substantially harder set of constraints. These constraints concern strong facial topology, bilateral symmetry expectations, and identity-consistency requirements that do not apply to arbitrary scene completion. This is because humans are hypersensitive to even slight visual violations in a way that does not apply to natural scenes. Building on these foundations, the field has progressed from CNN-based approaches to GAN-based methods, and now to the current transformer-based generation as surveyed by Elharrouss et al. [17]. HINT addresses two specific failure modes in prior methods: information loss during downsampling of corrupted images through mask-aware pixel-shuffle downsampling module (MPD), and inefficient global modelling by using Spatially-activated Channel Attention Layer (SCAL). HINT outperformed prior leading methods on the CelebA-HQ dataset by 3.3-5.7% in Peak Signal-to-Noise Ratio (PSNR) across mask sizes [18]. Bhattacharya et al. [19] take a different architectural route, a two-stage GAN combining CNNs for local features and vision transformers for global semantic layout, followed by multi-scale texture refinement. Despite the technical strengths of both approaches, both are trained and evaluated exclusively on CelebA-HQ and FFHQ. Analysis of the methods surveyed by Elharrouss et al. [17] reveals that approximately 85% evaluate on CelebA-HQ [17]. None of these benchmarks reflect infant facial geometry or the cleft-specific reconstruction task this project addresses.

Transformer-based approaches are especially well-suited for facial inpainting because of their self-attention mechanisms, which model long-range dependencies across the entire image. This allows transformers to capture global facial structure that CNN-based methods cannot. CNNs are constrained by local receptive fields, which causes them to struggle with inpainting larger missing areas that require distant context to infer a plausible completion. ICT [20] was an early and influential transformer-based pluralistic inpainting method that demonstrated that transformers could generate diverse and coherent completions by pairing a transformer for global structure with a CNN for local texture refinement. This hybrid architecture was a productive direction and set a strong early benchmark for the family. ITrans [21] extended this direction with a dual transformer design, a global transformer aggregating high-level context and a local transformer extracting low-level neighbourhood detail. They were able to demonstrate measurable improvements over purely CNN-driven approaches. However, like ICT, ITrans relies on downsampling operations that introduce information loss in corrupted image regions. MATdiff [22] integrates a mask-aware transformer as a conditional encoder with a diffusion model for generation, and reports competitive results. However, the authors acknowledge that the stochastic nature of diffusion generation produces semantic inconsistencies between known and inpainted regions. For cleft-lip reconstruction, the unpredictability of diffusion models is a noteworthy limitation rather than a benchmark caveat, given that the predicted output has clinical implications and must be structurally plausible. PUT directly handles the information loss problem that affects both ICT and ITrans through a patch-based autoencoder that avoids spatial downsampling entirely, and an unquantised transformer that operates on continuous features rather than discrete tokens. This produces measurable advances over previous transformer methods on complex datasets, and PUT reported improved performance on pluralistic inpainting tasks. PUT represents the most direct solution to the failure modes identified across the transformer family. Beyond its technical results, PUT is publicly available with

its code and pre-trained weights. This is a non-trivial consideration as across transformer-based inpainting methods, the gap between reported benchmark performance and practical retraining feasibility is significant. Several architectures with strong published results do not have complete code, have unclear training instructions, or demand computing resources that make domain adaptation prohibitive. PUT's accessibility therefore reinforces rather than substitutes for its technical selection. PUT is both the strongest technical candidate and the only one that can be realistically retrained on a new domain.

Facial image inpainting has advanced across both GAN and transformer architectures. HINT and Bhattacharya et al. [19] achieve a strong performance while transformer-based approaches and PUT in particular represent the most suitable direction for domain adaption to infant-specific data. Despite the progress, the literature across both model families shares the fundamental constraint that they are designed, trained, and evaluated on adult face datasets such as CelebA-HQ and FFHQ rather than infant-specific datasets. The combination of the difficulty of retraining inpainting transformer models on new domains, and the absence of any publicly available infant-specific facial dataset explains why no publicly available models currently exist for infant-specific cleft-lip reconstruction. This is not a simple gap in the literature but rather a consequence of two significant barriers that this project addresses. This gap requires a model that is capable of domain adaptation, but also the infant-specific training data that does not yet publicly exist, making synthetic data generation the critical next consideration.

### 2.3 Synthetic Infant Face Generation and Style-Based Domain Adaptation

In the field of medical imaging and computer vision, synthetic data generation has emerged as a recognised and principled strategy for tackling data scarcity and ethical constraints on real data collection [23]. Wang et al. found that synthetic augmentation has been shown to produce quantitative results gains across multiple clinical tasks, with recorded improvements of 12%, 13%, 17% on ophthalmic, chest, brain and breast imaging tasks respectively, particularly where real data is scarce or involves underrepresented populations [24]. A major reason why synthetic augmentation is being adopted is the ethical complexity of collecting real patient data, where privacy preservation remains a central concern. Generating synthetic data provides a valuable means of assembling training datasets that avoid any direct correspondence to real individuals [25]. This challenge is especially important for the paediatric population, where limited data availability, developmental variability, and ethical considerations have been identified as persistent barriers to AI development in child healthcare [26]. A recent systematic review of 181 public medical imaging datasets found that children represent less than 1% of available data, with only 1 of the 30 most cited datasets being paediatric, and only 17% of FDA-approved medical AI devices labelled for paediatric use [27]. The consequence of the underrepresentation is that adult-trained models have shown significant age bias when applied to paediatric populations, with measurable performance degradation in younger children [27]. Within the context of this project, generating synthetic infant facial data becomes a feasible and motivated response to a well-documented gap, providing training data that does not ethically exist in the public domain while avoiding the domain mismatch that adult datasets introduce.

Style transfer is a deep learning technique that combines the visual style of one image with the content of another, enabling the resulting image to look modified while maintaining the underlying structure. Gatys et al. introduced a neural algorithm of artistic style that showed how convolutional neural networks could separate and recombine the content and style of natural images [28]. Gatys et al.'s approach used feature representation from a pretrained CNN to define content loss and style loss, allowing an image to be optimised so that its high-level content matched one source while its texture statistics matched another [28]. Since its introduction, style transfer has moved significantly beyond purely artistic uses, evolving into a larger set of techniques for visual manipulation, domain transfer and feature-level adaptation across image distributions. This evolution is significant because it grounds style transfer as more than a visual effects tool but as a conceptual framework for transferring learned representations of appearance between domains [29]. This includes domain adaptation across age groups, demographics or visual conditions [29]. With style transfer providing the conceptual basis for the synthetic infant facial data generation, it motivates more specific literature on age transformation and child face synthesis.

Building on these conceptual foundations of style transfer, a large amount of work has been using these techniques for facial age transformation, where the goal is to modify a face image to look different at different ages while preserving identity. An example of this direction is the proposed model by Alaluf et al, the Style-based Age Manipulation (SAM) model. SAM formulates age transformation as a style-based regression problem within the latent space of a pretrained StyleGAN [12]. Instead of relying on discrete age groups or explicit ageing labels, SAM learns a continuous mapping from input images to age-conditioned latent codes. This enables the generation of plausible age-progressed or age-regressed faces from a single source image with strong identity preservation [12]. A complementary direction has focused specifically on generating synthetic child faces rather than translating existing adult images. ChildGAN proposed by Farooq et al., uses StyleGAN2 with domain adaptation to generate large-scale synthetic child facial data. This demonstrates that high-quality child face datasets can be created from limited real data while preserving privacy [30]. More recently, ChildDiffusion has extended this work using diffusion-based generation instead of GAN-based architectures, using stable diffusion and large language models (LLMs) to allow controllable augmentation and customisation of child facial data with text-conditioned attributes [31]. While these methods give confident evidence that synthetic child face generation is feasible, these models are not explicitly designed for infants, as infants (aged 0-2) are the age group on which cleft-lip reconstructive surgery is typically performed on. Beyond this, broad age translation and general child face synthesis are not equivalent to the infant-specific anatomical realism required for cleft lip out-

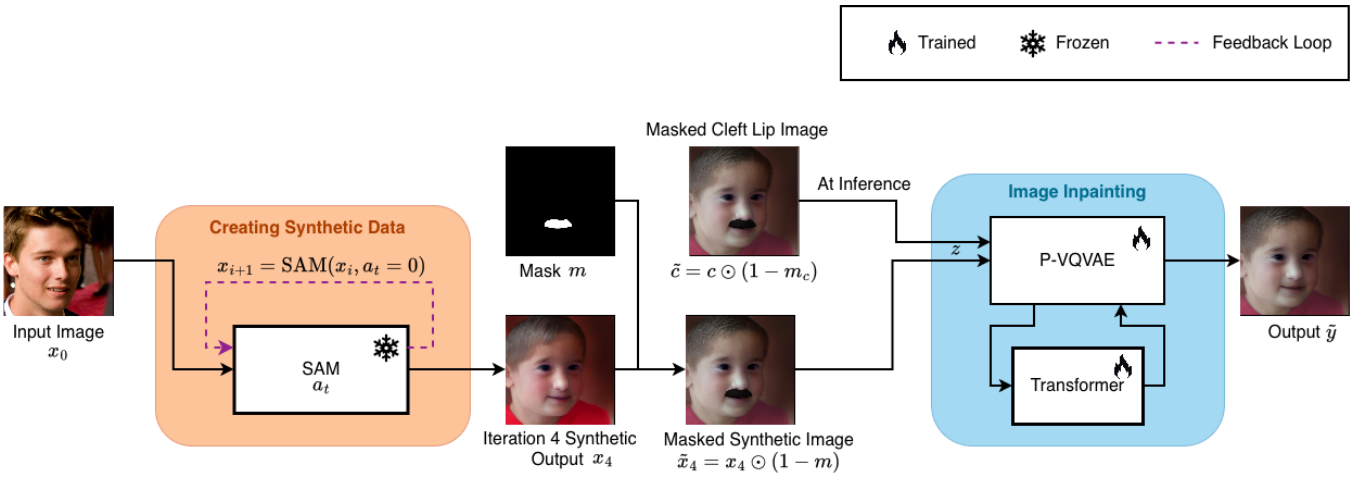



Fig. 2: Overview of the infant-specific cleft lip inpainting pipeline. Inpainting model starts with processing the input image of an adult face $x_0$ with the pretrained SAM age regression model at target age $a_t = 0$ to generate a sequence using the iterative feedback loop whereby $x_{i+1} = \text{SAM}(x_i, a_t = 0)$ for four iterations resulting in a final output infant image $x_4$. This image along with a randomly generated binary mask $m$ is then used to generate a masked image of the infant face $\tilde{x}_4 = x_4 \odot (1 - m)$ that acts as the input to the PUT inpainting model. The pretrained PUT model consists of a Patch-Based Variational Autoencoder (P-VQVAE) model and an Unquantised Transformer (UQ-Transformer). Both of these models have been retrained using the synthetically generated infant dataset. In the case of the actual cleft lip image $c$, a combination of this and the mask yields $\tilde{c} = c \odot (1 - m_c)$ which is fed into the PUT model during inference to obtain the output $\tilde{y}$ – a non-cleft infant image.

come prediction. This is where features such as philtrum, lip vermillion, and nasal base must be reproduced with high spatial fidelity. These works are therefore adjacent to, rather than direct solutions for the infant-specific synthetic data generation that this project requires. However, they collectively establish that style-transfer-based approaches to age and demographic translation can produce realistic, identity-preserving facial outputs at scale.

Taken together, there are three established areas: cleft lip image generation, which has been shown to be clinically useful but is limited by small and private data; modern inpainting architectures, proving capable but trained predominantly on adult faces; and style-transfer-based approaches to age translation, which have matured to the point where synthetic facial data can be generated at scale. Chen et al.'s feasibility study, the PUT transformer-based model, and style-transfer-based age-transformation methods like SAM each represent the strongest current work in their respective areas [12], [13], [14]. However, no current work combines these three directions into a single pipeline that generates synthetic infant facial data, retrains a transformer-based inpainting model on that data, and applies the result to cleft lip surgical outcome prediction. The outcome of cleft lip surgery requires anatomically faithful infant facial reconstruction, which cannot be achieved by adult-trained models due to domain shift, but also cannot be simply addressed by retraining on infant data due to ethical and practical constraints on those datasets. Resolving this gap motivates the methodology presented in the following section, which combines style-transfer-based synthetic infant data generation with a retrained transformer-based inpainting model to enable infant-specific prediction of cleft lip outcomes.

## 3 Methodology

In this section, we present the overview pipeline, followed by the methodology to create the synthetic infant dataset, giving an explanation of why CelebA-HQ was chosen as the primary dataset and how SAM works and is utilised. After this, we describe how the PUT model works and how we retrained it. This section ends with detailing the training configuration and the experimentation conducted that led to this final pipeline.

### 3.1 Pipeline Overview

The proposed method for this research includes developing the inpainting pipeline aimed at the infant. The mentioned process comprises two steps: generating images of infant faces and training the PUT transformer-based inpainting model. During inference, when the real cleft lip facial image is used as the input, the facial image itself serves as the input to the PUT model. Such decisions are made because the facial image does not require age regression since it is the image of the infant itself.

The training pipeline, shown in Figure 2, begins with adult facial images from the CelebA-HQ dataset. Each adult image from the dataset is passed through SAM iteratively for four iterations in a feedback loop, where each iteration uses the previous output as its input, with the target age being set to 0 years old. This process produces a synthetic dataset of 30,000 images of 256x256 resolution. Each synthetic image is paired with a randomly generated binary mask to create a masked synthetic image for training. PUT contains two major components, P-VQVAE and UQ-Transformer. We retrain these components on the masked synthetic images from the existing weights.

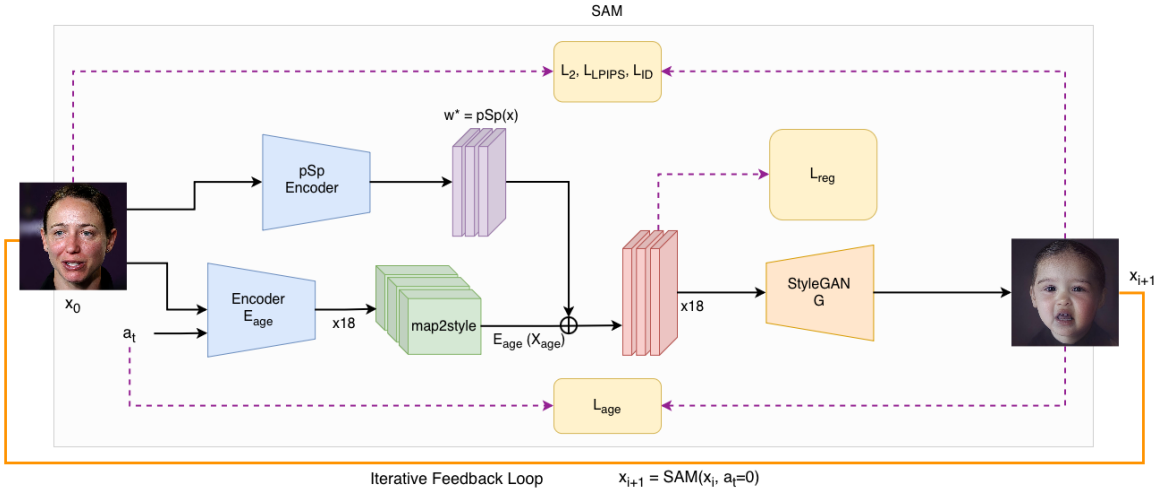



Fig. 3: Iterative SAM feedback loop used to generate synthetic infant faces. At each iteration, the output $x_{i+1} = SAM(x_i, a_t = 0)$ is fed back as the input to the next pass, with the target age fixed at $a_t = 0$ throughout. All components are pretrained.

At inference, the trained PUT model receives a masked cleft lip image and inpaints the concealed region, as seen in Figure 2.

### 3.2 Synthetic Infant Dataset Generation

The synthetic infant dataset is generated from CelebA-HQ, a publicly available high-quality facial image dataset consisting of 30,000 adult face images at 1024x1024 resolution [32]. CelebA-HQ was chosen as the source dataset because it provides 30,000 images, which is sufficiently large to train an inpainting model. Additionally, it consists only of adult faces which is the correct source domain for SAM since SAM is designed to take adult face inputs and transform them towards a target age. CelebA-HQ is openly available for research use without requiring extra ethical approval. As a dataset sourced from celebrity photographs, CelebA-HQ does not give a balanced demographic representation. It skews towards lighter-skinned subjects and younger adults, this is a known characteristic of celebrity-sourced datasets [33].

To generate synthetic infant faces from this dataset, each image in CelebA-HQ is processed using SAM. SAM is a style-based age regression model that performs continuous age transformation by operating within the $W+$ latent space of a pretrained StyleGAN2 generator [12]. SAM is used entirely as a frozen pretrained component and is not retrained in this project. The core of SAM's forward pass is the ageing encoder $E_{age}$ , which takes both the input face image $x_0$ and the target age $a_t$ and extracts feature maps at three different spatial scales. These feature maps are passed through 18 map2style blocks that progressively downsample them into 18 512-dimensional style vectors, encoding the age-conditioned transformation into the $W+$ latent space as $E_{age}(X_{age})$ [12]. In parallel, a frozen pretrained pSp encoder extracts the $W+$ latent code $w^*$ of the input image $x_0$, representing the person's identity independently of age [34]. This identity code $w^*$ is combined with the age-conditioned style vectors from $E_{age}$ through element-wise addition, creating an aggregated latent code that simultaneously encodes both the desired age transformation and the original facial identity [12]. The aggregated latent code is passed into the frozen pretrained StyleGAN generator $G$, this decodes it into a photorealistic output image $x_{i+1}$ which represents the face output and is the input face transformed towards the target age while preserving the identity of the input [12]. During SAM's original training, visual similarity losses L2 and LPIPS, identity loss LID, regularisation loss $L_{reg}$, and age guidance loss $L_{age}$ together ensure the output is visually plausible, identity-preserving, and correctly aged. These are all computed using fixed pre-trained networks. Prior to each inference pass, SAM's pipeline applies face alignment using dlib's 68-point facial landmark predictor to standardise head pose and facial position. This is followed by resizing the aligned image to 256x256 resolution. These preprocessing steps are applied each time an image is passed through SAM, ensuring each pass receives a consistently standardised input.

In this project, rather than applying SAM once, we apply it multiple times to generate the final synthetic dataset through an iterative feedback loop in which each output of SAM's pass is fed back as the input to the next iteration. The iterative loops keep a fixed target age at $a_t = 0$ throughout. This process can be seen in Figure 3. The process is defined by the recurrence relation $x_{i+1} = SAM(x_i, a_t = 0)$ for i $\in$ 0, 1, 2, 3, where $x_0$ is the original CelebA-HQ image and $x_4$ is the final synthetic infant face used for training. SAM is applied multiple times in sequence because the transformation magnitude achievable in a single forward pass is constrained by the geometry of the $W+$ latent space [12]. A single application of SAM produces a younger-looking face but does not achieve sufficiently infant-like results. Each successive pass re-encodes the transformed face as a new input, allowing the pSp encoder to extract a new identity code and the aging encoder to apply a fresh age-conditioned shift from the current facial state [12]. Across iterations, the face progressively develops infant-like characteristics such as softer skin texture, rounder facial features, and reduced facial bone definition. The number of iterations was

determined through qualitative experimentation across 1 to 5 iterations. At five iterations, skin texture degraded toward aged rather than infant-like characteristics, the opposite of the desired transformation, consistent with observed GAN behaviour at latent space extremes. Four iterations produced the most infant-like results while maintaining facial plausibility (with quantitative FID support from Table 2 discussed in Section 5). Applying this process to all 30,000 images in CelebA-HQ produced the synthetic infant dataset $Dsynth = x_4(j)$ $j$ = 1 to 30,000, where each image is a 256x256 synthetic infant face that does not correspond to any real individual. This dataset forms the training set used to retrain the PUT inpainting model.

### 3.3 Image Inpainting

To perform image inpainting, PUT was used for the reasons mentioned in section 2.2. This included its patch-based architecture, which avoids information loss during downsampling, strong performance on pluralistic inpainting tasks, and its publicly available pretrained weights and codebase, which make domain adaptation through retraining feasible [13]. In this project, PUT is retrained on the synthetic infant dataset described in Section 3.2 to adapt it from its original general-purpose training domain to the infant facial domain, enabling it to perform plausible inpainting on infant faces.

PUT consists of two sequentially dependent components, the Patch-based Vector Quantised Variational Autoencoder (P-VQVAE) and the Un-quantised Transformer (UQ-Transformer). Each component addresses a specific source of information loss that affected prior transformer-based inpainting methods. Transformer inpainting methods before PUT had two main sources of information loss, spatial downsampling of the input to reduce the computational cost and quantisation of pixel values into a small discrete codebook, which discards colour precision [20]. PUT addresses these through its two-component architecture. The P-VQVAE handles the downsampling problem by dividing the input image into 1,024 non-overlapping patches by using a kernel with a stride of 8 and processing each independently through a patch encoder, avoiding spatial downsampling while maintaining computational efficiency. Each patch is encoded into a feature vector and quantised using either a codebook for unmasked patches that represents visible image content, or a separate codebook for masked patches representing regions to be inpainted. Then the decoder reconstructs pixel values from these quantised representations, keeping unmasked regions unchanged while synthesising content for the masked regions. The UQ-Transformer addresses the quantisation problem by taking the raw unquantised feature vectors from the P-VQVAE encoder as input rather than discretised tokens. This allows the model to preserve more precise information during the prediction stage. Using self-attention layers across all 1,024 patch positions simultaneously, the transformer predicts which codebook entry is most similar to each masked patch given the global context of all visible patches. The transformer uses only quantised tokens as prediction targets instead of inputs, ultimately combining the precision of continuous features with the structured output space of the codebook. In this project, the architecture of PUT is used in the same way as proposed by Liu et al. [13], with no structural changes made to either component. We retrain PUT's weights on the infant domain rather than any architectural change.

Retraining PUT on the synthetic infant dataset follows a two staged sequential process that mirrors the original training procedure, where P-VQVAE is retrained first, and then UQ-Transformer is retrained using the frozen pre-trained P-VQVAE as its encoder backbone. In the first stage, the P-VQVAE is retrained on the synthetic infant training set initialised from PUT's existing pretrained weights rather than beginning from scratch due to the difference between the distribution of the datasets Places2 and ImageNet compared to our synthetic infant dataset. The model learns to encode and decode infant facial patches, adapting its codebook representations to the visual characteristics of synthetic infant faces. This includes softer skin texture, rounder facial geometry, and different eye-to-face proportions compared to the adult faces it was originally trained on. In the second stage, the retrained P-VQVAE is frozen and its weights are loaded as a fixed encoder backbone for the UQ-Transformer. The transformer is retrained on the same synthetic infant dataset, using the configuration in Table 1, learning to predict infant-domain codebook tokens from the infant-specific feature representations produced by the retrained encoder. The two-stage approach is necessary because the UQ-Transformer's predictions are based in the P-VQVAE codebook such that, if the transformer were trained before the codebook had been adapted to the infant domain, the predictions would be anchored to adult-domain representations and the domain adaptation would be incomplete. Both the P-VQVAE encoder and decoder are retrained in stage 1. In stage 2, the full P-VQVAE is frozen and only the UQ-Transformer weights are updated. Ultimately adapting PUT to the infant facial domain.

The synthetic dataset used for retraining was limited to 10,000 images, where 8,000 were used for training and 2,000 for testing. We do not use the complete 30,000 image dataset due to computational constraints, specifically the memory and time limitations of the available HPC resources on the Durham University's NCC cluster. While a larger training set would likely improve performance, 8,000 images represents a practical upper bound given the available resources and the substantial training time required for both stages. We additionally retrain HINT [18] on the same dataset as a transformer baseline.

### 3.4 Training Masks and Cleft Lip Inference

During training, each synthetic infant image $x_4$ is paired with a randomly generated binary mask $m$, producing a masked synthetic image $\tilde{x}_4 = x_4 \odot (1 - m)$ which serves as the input to the PUT inpainting model (Figure 2). Masks are generated randomly with image coverage ranging from 20% to 60%, covering irregular regions of the face. The coverage range of 20% to 60% was chosen to ensure a meaningful inpainting challenge whilst maintaining enough surrounding facial context for the model to predict a reasonable reconstruction. Masking less than 20% of the image would leave insufficient training signals whereas masking

TABLE 1: Training Configuration for P-VQVAE and UQ-Transformer Retraining

| Parameter | P-VQVAE | UQ-Transformer |
|---|---|---|
| Optimiser | Adam ($\beta_1$=0.0, $\beta_2$=0.9) | AdamW ($\beta_1$=0.9, $\beta_2$=0.999) |
| Peak learning rate | $2 \times 10^{-4}$ | $1.5 \times 10^{-3}$ |
| Minimum learning rate | $1 \times 10^{-6}$ | $1 \times 10^{-5}$ |
| LR schedule | Cosine annealing | Cosine annealing |
| Warmup steps | 3,000 | 28,750 (∼20 epochs) |
| Batch size | 8 | 6 |
| Max epochs | 150 | 300 |
| Image resolution | $256 \times 256$ | $256 \times 256$ |
| Training images | 8,000 | 8,000 |
| Test images | 2,000 | 2,000 |
| Hardware | NVIDIA RTX A6000 | NVIDIA Ampere GPU |
| Approx. training time | ∼3 days | ∼1 day |

more than 60% of the image would remove too much facial context for reliable inference. No anatomical guidance is used in mask placement during training. Random masks are used during training to ensure the model can do robust inpainting across a wide variety of missing region sizes and locations, rather than overfitting to a specific mask shape or position. This is important as cleft lip infants may have differently shaped cleft lips such that masks placed over different cleft lip images will vary in shape and coverage, reflecting the natural variation in cleft representation.

In contrast to the training process, during inference, the input is an image $c$ of a cleft lip infant image. SAM is not needed as the input should already be an infant face where age regression is not required. Then each mask $m_c$ is placed over the cleft region of each input image to conceal it, therefore producing the masked cleft lip input $\tilde{c} = c \odot (1 - m_c)$. The mask is sized to cover the cleft and surrounding areas to allow the model to have sufficient context to predict a plausible reconstruction (Figure 2). The masked cleft lip image is then passed into the retrained PUT model, this inpaints the concealed region to create the post-surgical predicted facial appearance, removing the cleft lip.

## 3.5 Experimentation

### 3.5.1 Neural Style Transfer

Beginning with the neural style transfer as described by Gatys et al., we use CelebA-HQ, where the adult face is the content image and an infant image from UTKFace infant image aged 2 or under for the style image [35]. The expected outcome was that the adult face would gain the infant-like texture and surface appearance of the infant image while retaining the underlying facial structure of the adult image. However, as shown in Figure 4, the output was visually nearly identical to the input adult face. The resulting image displayed slightly smoother skin and underwent no meaningful structural or appearance change toward infant characteristics. The output was consistent with how neural style transfer operates, optimising for matching Gram matrix texture statistics between the style and content images and not modelling geometric or structural changes [28]. Because infant faces have significantly different facial geometry and proportional structure compared to adult faces, such as head shape and size, style transfer does not fully capture the most important dimension of the infant-adult domain gap. This confirms that texture-level manipulation alone is insufficient for synthetic infant generation and that geometric transformation of facial structure is required in future approaches.

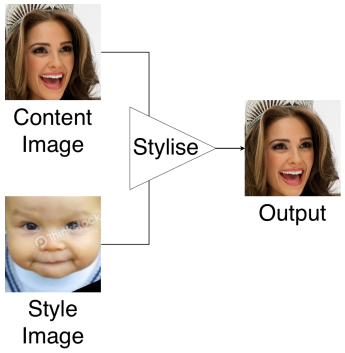



Fig. 4: Neural style transfer applied to a CelebA-HQ adult face (content image) using a UTKFace infant face (style image), with Right image being the stylised output image.

### 3.5.2 Landmark Warping with Style Transfer

As neural style transfer does not do the geometric transformation required to convincingly turn an adult image into an infant, a two-stage pipeline was developed that combines facial landmark warping with neural style transfer. This pipeline aims to reshape adult facial structure towards infant proportions before applying texture-level style adaptation. In the first stage, facial landmarks were detected on both the adult and infant source images. Adult landmarks are then geometrically transformed to match the placement of the infant landmark positions, reshaping the adult face towards infant proportions. Using the dlib Python library, we initially detect 68 facial landmarks [36]. As seen in the top row of Figure 5, geometric warping towards infant proportions is visible, the warping introduces significant visual distortions and noise, particularly around the nose and mouth region, producing incorrectly proportioned results. To improve warping quality, the landmark count was increased to 468 using MediaPipe face mesh detection [37]. As seen in the bottom row of Figure 5, this resulted in substantially more natural geometric deformation with noticeably fewer distortions compared to the 68 landmark approach. This demonstrates that denser landmark coverage does better capture the fine-grained structural differences

between adult and infant images. Despite these improvements, both landmark configurations produced outputs that retained adult hair, adult skin characteristics and adult facial lines. The approach required a paired infant reference for each transformation, making it impractical to scale 1-to-1 for 30,000 images without a sufficiently large dataset or using testing one-shot learning. These findings revealed that geometric warping improves upon the original style transfer but remains insufficient on its own, and that a generative approach capable of simultaneously modelling both structural and appearance changes would be required.

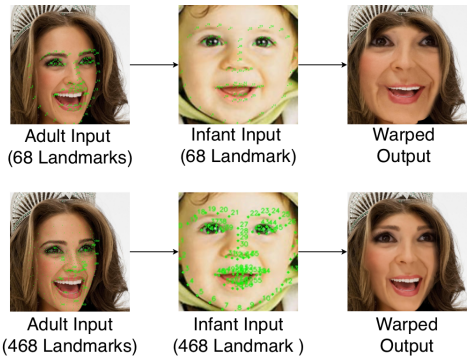



Fig. 5: Facial landmark warping combined with neural style transfer. Top: 68-point dlib landmarks; bottom: 468-point MediaPipe landmarks. Left: adult input; centre: warped face; right: style transfer output.

### 3.5.3 Diffusion Model with LoRA fine-tuning

To overcome the limitation of landmark warping alone, specifically the failure to create convincing infant skin textures and appearances, the warped outputs are then passed into a generative Stable Diffusion 1.5 fine-tuned with Low rank adaptation (LoRA) on the UTKFace 0-2 year old infant subset of 1,605 images [35], [38], [39]. LoRA allows the generative model to be conditioned towards the infant domain. We tested both the 68 and 468 landmark warped inputs through the same fine-tuned diffusion model. Figure 6 shows representative outputs from both variants. The 468 landmark variant produced cleaner outputs than the 68 landmark variant, which displayed visual noise and texture artefacts. However, both variants shared persistent visual issues, including artefacts around the nose region, inconsistent iris rendering, adult facial lines remaining, and outputs that appear to be young children (approximately 3-5 years old) rather than 0-2 years old. These results could be attributed to there being 1,605 training images. This is insufficient to fully condition a diffusion model on the infant domain, this is consistent with the data scarcity problem for infant images mentioned in the related work section [27]. Additionally, the diffusion generation process does not explicitly preserve the identity of the input face, meaning the warped geometry provides only structural guidance, while the generative model still introduces inconsistencies that undermine the output's plausibility. Despite demonstrating greater structural flexibility than traditional style transfer, this approach was ultimately unable to produce realistic infant faces due to limited training data, inconsistent identities, and persistent visual artefacts. These findings confirmed that a generative model with explicit age control and identity preservation was needed, motivating the exploration of age regression approaches in the following experiment.

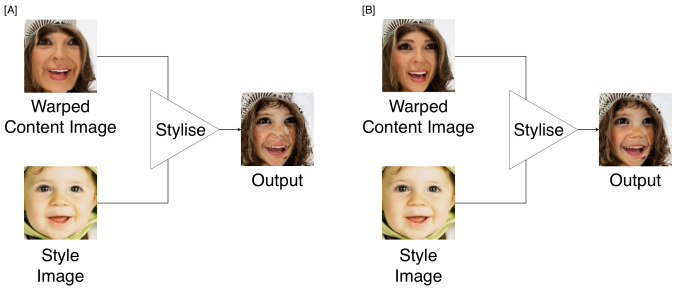



Fig. 6: Stable Diffusion 1.5 with LoRA fine-tuned on UTKFace (0–2 years) applied to landmark-warped adult faces. (A) 68-point warping; (B) 468-point warping. Top: warped content; bottom: style image; right: diffusion output.

### 3.5.4 SAM Target Age

The limitations of style transfer and diffusion-based approaches established that infant face generation requires a model which is able to have explicit age control and structural facial transformation while preserving identity. These limitations motivated exploring SAM, which operates directly in the $W+$ latent space of StyleGAN2 and performs continuous age regression rather than texture or geometry manipulation [12]. To identify the optimal target-age configuration, SAM was applied to a selection of CelebA-HQ images in a single forward pass, with target ages set to 0, 1, 2, 3, 4, and 5 in separate runs. Figure 7 shows two representative examples across this age range. Decreasing the target age produced marginally younger-looking facial appearances in terms of rounder features and softer skin texture. The visual differences across the range of 0 to 5 were subtle and difficult to distinguish in a single pass, with all outputs appearing to represent young children rather than infants. Target age 0 was selected as the fixed target age on the basis that it represents the youngest possible setting and therefore gives the strongest directional signal towards infant appearances within the model's latent space. The transformation magnitude achievable in a single pass is constrained by the geometry of the W+ latent space, and a single application was insufficient to reach the infant age regardless of the target age setting [12]. Despite this limitation, SAM produced results that were considerably more convincing than those from traditional style transfer and diffusion methods. SAM produced outputs free of visual artefacts and preserved identity whilst applying appropriate geometric transformations to the face structure. The approach was also scalable to the full 30,000 image CelebA-HQ dataset without requiring a paired infant reference image for each transformation. These findings motivated the iterative feedback loop described in Section 3.2, in which SAM is applied multiple times in sequence to progressively push the transformation deeper toward the infant age range than a single pass allows.

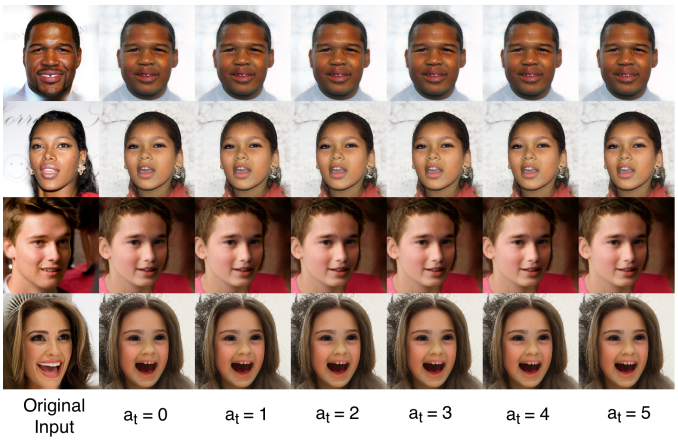



Fig. 7: Single SAM forward pass on a CelebA-HQ adult face across target ages 0–5. Left to right: original input, then outputs at ages 0, 1, 2, 3, 4, and 5.

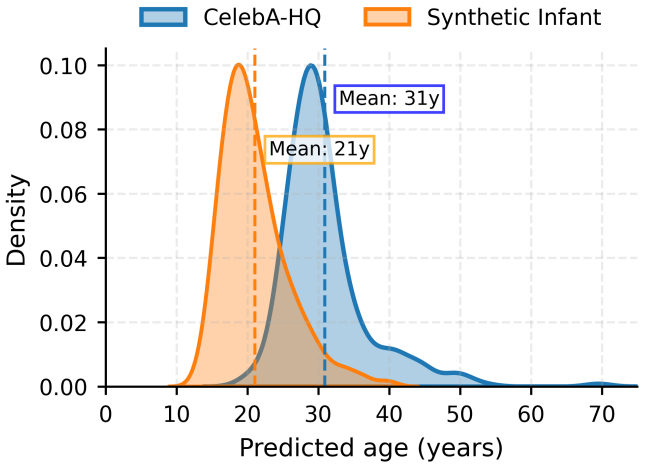



Fig. 8: Age distribution of predicted ages for Iteration 4 of the synthetic infant dataset compared to CelebA-HQ, estimated using the DEX age prediction model.

## 4 Results

### 4.1 Implementation Details

The full implementation details for this project including training configuration, dataset splits, and hardware are described in Section 3. Synthetic dataset evaluations use UTKFace filtered to ages 0-5 as the reference distribution for FID comparison. For the inpainting evaluation, we validate the retrained model using 2,000 unseen images from the synthetic infant dataset. The synthetic dataset quality is assessed using FID measured against both the reference dataset and CelebA-HQ. The inpainting quality is assessed using PSNR, SSIM, and LPIPS computed against the original unmasked ground truth images, chosen to capture pixel-level, structural, and perceptual quality respectively.

### 4.2 Synthetic Dataset Results

In Figure 9 we present six representative examples of the synthetic infant face generation process, showing the CelebA-HQ adult input alongside the corresponding synthetic infant output produced after four iterative SAM passes with target age $a_t = 0$. All 6 examples produce visually younger-looking faces with softer skin texture, rounder facial features, and reduced bone definition compared to the adult inputs with consistent results observed across subjects of different demographics including subjects of different genders and ethnicities. In Table 2, the FID scores for each iteration of the synthetic dataset are compared against the UTKFace reference dataset and against CelebA-HQ. The FID against the reference dataset decreases from 195.48 for CelebA-HQ to 68.81 at iteration 3, before rising to 70.24 at iteration 4 and 72.50 at iteration 5. FID against CelebA-HQ increases monotonically across iterations from 0.0 to 213.48 at iteration 5. Iteration 4 was selected as the final output for the synthetic dataset as described in Section 3.5, despite iteration 3 reaching a marginally lower FID against the reference dataset.

Figure 8 presents the predicted age distributions of CelebA-HQ and the synthetic infant dataset at iteration 4, estimated using the DEX age predictor. The CelebA-HQ distribution shows a mean at 31 years, reflecting the adult population of the dataset. The synthetic infant dataset shows a mean of 21 years, representing a shift towards younger predicted ages by 10 years. The synthetic distribution is narrower than the CelebA-HQ distribution with less spread toward older age ranges.

TABLE 2: FID Scores for Synthetic Dataset Evaluation

| Dataset | FID vs Reference | FID vs CelebA-HQ |
|---|---|---|
| CelebA-HQ (baseline) | 195.48 | 0.0 |
| Iteration 1 | 98.93 | 83.07 |
| Iteration 2 | 74.18 | 132.39 |
| Iteration 3 | 68.81 | 172.41 |
| Iteration 4 | 70.24 | 197.71 |
| Iteration 5 | 72.50 | 213.48 |

Table 3 presents a metadata comparison of our synthetic dataset against commonly used facial datasets in computer vision tasks. This table highlights the infant data scarcity problem by showing the proportion of images of infants within each dataset. CelebA-HQ has no infant images. FFHQ, UTKFace and IMDB-WIKI contain very few, with IMDB-WIKI yielding just 409 infant images out of its 520,666 total images (0.08%). These datasets were not collected for infant-specific tasks. Whereas IFEPI contains 25,000 infant images but is unsuitable as the images are of close-up pain expressions only framing 120 unique subjects [40]. Our dataset contains 30,000 full face images specifically targeting the infant domain with no duplicates and no ethical concerns around real infant data.

TABLE 3: Comparison of Publicly Available Face Datasets

| Dataset | Total | Infant (0–2) | Type | Framing |
|---|---|---|---|---|
| Synthetic Infant (Ours) | 30,000 | 30,000 (target) | Synthetic | Full face |
| CelebA-HQ | 30,000 | 0 | Real | Full face |
| FFHQ | 70,000 | 2,492 | Real | Full face |
| UTKFace | 23,708 | 1,605 | Real | Full face |
| IMDB-WIKI | 520,666 | 409 | Real | Full face |
| IFEPI | 25,000 | 25,000 | Real | Close-up |




Fig. 9: Six examples of synthetic infant face generation from CelebA-HQ adult inputs. Top row shows the original adult input images. Bottom row shows corresponding synthetic infant outputs produced after four iterative SAM passes with target age $\alpha_t = 0$.

### 4.3 Image Inpainting Results

Figure 10 presents the results from inpainting the synthetic infant test images using two different types of masks: randomly generated masks in Figure 10a and manually-drawn general cleft-area masks in Figure 10b. In Figure 10, both the input and the output results are presented for each configuration example. With the random mask, the model outputs reconstructions that are visually consistent with the context provided by the remaining unmasked parts. In the general cleft-area mask scenarios, the model produces reconstructions for the masked lips in each example. The inpainted cleft region shows subtle variation in the consistency across the six subjects, where the size of the upper lip varies from person to person and occasionally differs from the ground truth. Table 4 presents the quantitative evaluation results for the retrained PUT model on the synthetic infant test set, recording the PSNR, SSIM, and LPIPS scores of each model tested. The retrained PUT model reaches a PSNR of 29.86, an SSIM of 0.9536, and an LPIPS of 0.1057 on the synthetic infant test set. The pretrained PUT baseline achieves a PSNR of 30.22, an SSIM of 0.9581, and an LPIPS of 0.1164.

TABLE 4: Quantitative Inpainting Results

| Model | PSNR↑ | SSIM↑ | LPIPS↓ |
|---|---|---|---|
| Pretrained PUT | 30.22 | 0.9581 | 0.1164 |
| Retrained PUT (Ours) | 29.86 | 0.9536 | 0.1057 |
| Retrained HINT | 31.87 | 0.9500 | 0.1020 |

### 4.4 Cleft Lip Inference Results

In Figure 11, cleft lip inference examples are shown from images retrieved from publicly available online resources [41], [42]. The figure depicts how the masked input of the cleft lip is shown, along with the output image generated after applying the retrained PUT model to the problem statement, where the model predicts the expected post-surgical results. In this process, the model reconstructs the masked cleft region to produce a continuous lip structure that is consistent with the surrounding facial features. As there is no publicly available post-surgical ground truth for comparison, quantitative metrics are not computed for cleft lip inference. Therefore, evaluation for cleft lip inference is strictly qualitative based on visuals.

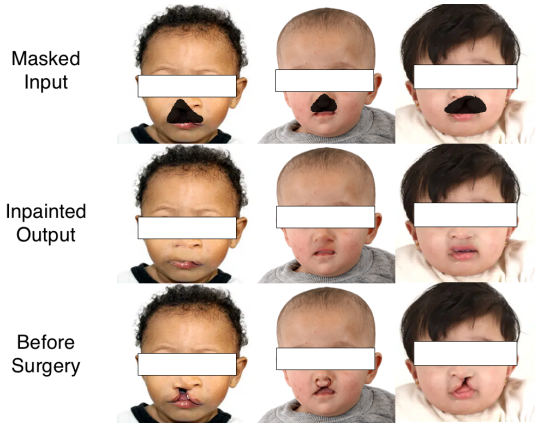
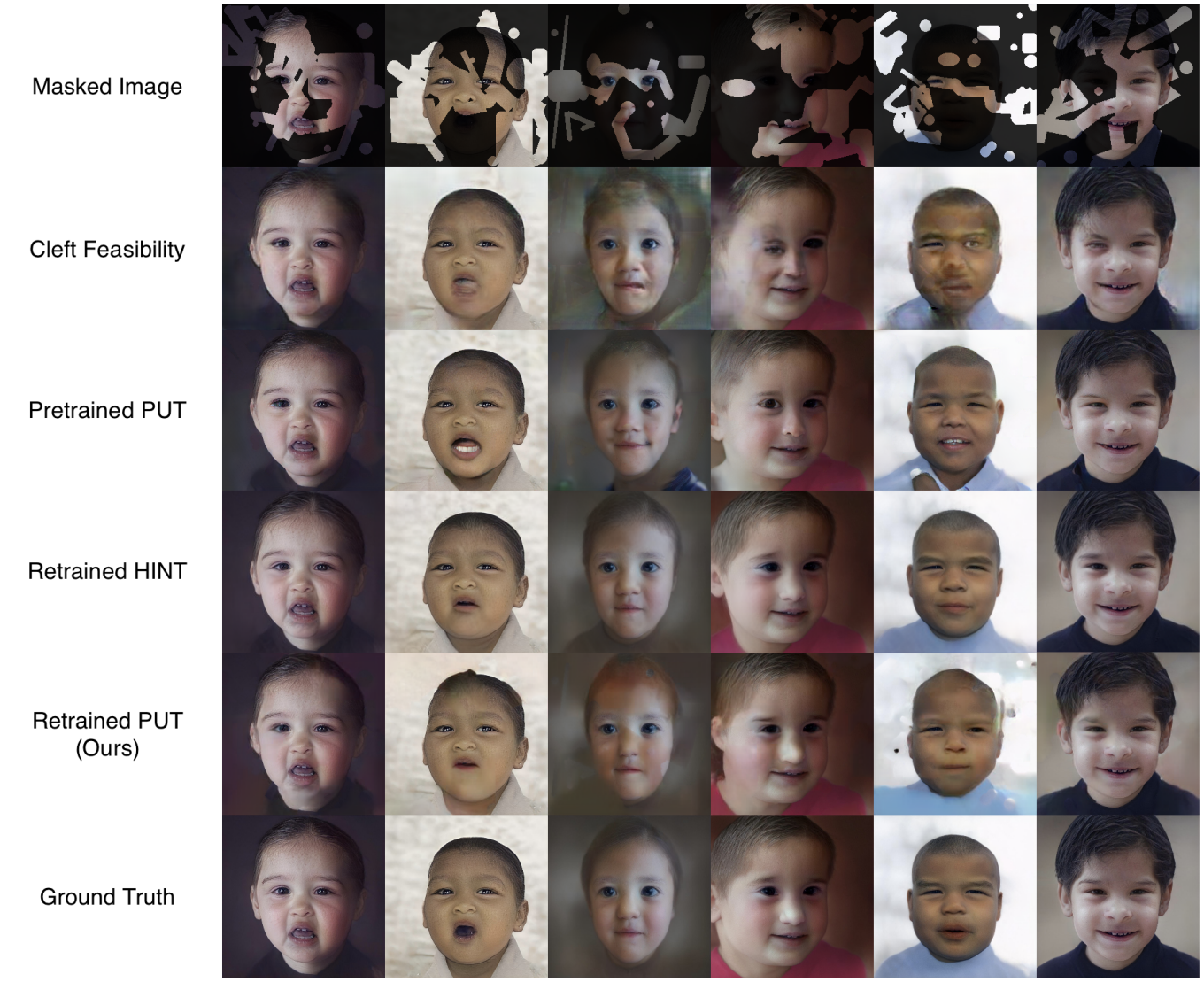



Fig. 11: Qualitative cleft lip inference results on real infant images. Each pair shows the masked cleft lip input and the corresponding inpainted output.

## 5 Evaluation

### 5.1 Synthetic Dataset Evaluation

As shown in Table 3, no existing publicly available dataset provides sufficient full-face infant images for training a generative model, with even large datasets like IMDB-WIKI, which contains only 409 images out of 520,666. On the other hand, IFEPI dataset contains 25,000 infant images on their close-up pain expressions with the no pain images being the most usable for our task. However, the images from this dataset are from only 120 unique subjects, making it unsuitable for full face inpainting tasks. These datasets illustrate that the challenge is not only the lack of infant images, but also the absence of suitable full-face infant data at scale. Our synthetic dataset was able to overcome these issues by having a size of 30,000 images which is large enough to train a generative model specifically targeting the infant domain. Additionally, each image corresponds to a unique synthetic individual with no duplicates, providing a diversity of faces and facial features required for training a robust generative model. The dataset also adheres to full-face framing throughout all images, being consistent with standard computer vision facial datasets.




(a) Qualitative inpainting results on synthetic cleft-lip examples. The masked image is shown in the first row, followed by the inpainted outputs of different models and the corresponding ground truth.




(b) Qualitative comparison between the masked image, the proposed inpainted outputs, and ground truth images.

Fig. 10: Qualitative evaluation of infant-face inpainting under clinically motivated and irregular mask settings.

#### 5.1.1 Quantitative Evaluation (FID)

FID is used to measure distributional similarity between dataset pairs [43]. To establish a baseline adult-to-child distributional distance, we calculate the FID of our input adult dataset, CelebA-HQ, against the children's reference dataset. Additionally, to measure how much the transformation step closed this gap, we calculate the synthetic dataset against the same children's reference dataset as well as the synthetic dataset against CelebA-HQ to confirm the transformation has moved images away from the adult distribution. As there are no complete publicly available infant datasets with high quality images [27], we use UTK-Face, filtered specifically for images of people aged 0-5 as it is closer semantically to infants than adult images [35]. UTKFace specifically has images which compared to other datasets, had poses and quality most similar to CelebA-HQ, our baseline. The decrease in FID against the reference dataset from 195.48 for CelebA-HQ to 68.81 at iteration 3 indicates that the synthetic images are substantially closer in distribution to the younger reference faces than the original adult inputs. Suggesting that the iterative SAM transformation successfully shifts the facial appearance towards younger age groups. Despite iteration 3 achieving the lowest FID against the reference dataset, iteration 4 was selected based on qualitative visual assessment as described in Section 3.5. This discrepancy highlights a known limitation of evaluating using FID, it measures the distributional similarity at the feature level but does not directly recognise the perceptual infant-likeness. This means that a slightly higher FID score does not strictly indicate a lower visual quality for the specific target domain. The monotonic increase in FID against CelebA-HQ across iterations supports the idea that each successive SAM pass moves the synthetic images progressively further from the adult distribution, consistent with the intended transformation. However, one limitation of this evaluation is that the UTKFace dataset was filtered to ages 0-5 rather than 0-2 specifically. This leaves the reference distribution to include toddlers and young children beyond the traditional infant age range targeted by this project. Ultimately this broadens the reference distribution and gives a potential explanation for why the FID scores do not reach lower values even at iteration 4. Nonetheless, the ages 0-5 were used such that there would be a reference dataset large enough to have a confident comparable distribution to use for comparison and capture the essence of younger looking faces.

#### 5.1.2 Age Distribution Analysis

A pretrained age predictor is applied to both CelebA-HQ and the synthetic dataset to estimate the predicted age distribution of each dataset, providing a complementary measure of whether the transformation generates faces that are perceived as younger by an estimation model. Figure 8 presents the predicted age distribution of CelebA-HQ with the two distinct peaks, one centered around adult ages for CelebA-HQ and one shifted towards lower ages for the synthetic dataset. However the synthetic infant images are predicted to have a mean of 21 years old rather than the target infant range of 0-2, which at first appears to be a significant limitation of the synthetic dataset. This result is likely attributable to several factors rather than a fundamental failure of the generation pipeline. The age predictor we use is DEX, which is trained on the IMDB-WIKI dataset and contains very few images of people 10 or younger [44]. This meant that their model struggles with predicting images that are actually in the infant age range because of the model's limited calibration on infant faces. This results in overestimation of ages at the lower ends of the distributions. Additionally, SAM does not explicitly model hair transformation as the generative process focuses on facial geometry and skin texture and does not consistently transform the adult hair. Several synthetic images retain long or styled adult hair which is visually inconsistent with infant appearances and likely influences the age predictor towards higher estimates. DEX uses the full image rather than cropping the specific facial region, meaning that the background elements, hair and clothing from the original adult source image may contribute to the age prediction alongside the transformed facial features [44]. Despite these limitations, the shift in the predicted age distribution between CelebA-HQ and the synthetic dataset confirms that the transformation produces faces that are consistently perceived as younger, even if the absolute predicted ages do not reach the infant range. This is consistent with the visual results shown in Figure 9 and the FID findings discussed above.

#### 5.1.3 Qualitative Assessment

Qualitative visual assessment of the synthetic dataset supports that the iterative SAM pipeline successfully creates visually convincing infant-like facial outputs across a diverse range of subjects. The six examples in Figure 9 span subjects of different genders and ethnicities and the transformation produces consistently infant-like results across all examples, demonstrating that the pipeline generalises across demographics in CelebA-HQ well. A notable strength of SAM-based approaches is that the identity-preserving transformation ensures each of the synthetic outputs remains visually distinct. This is achieved by carrying the facial structure of each unique CelebA-HQ source identity through to the synthetic output, inheriting the visual diversity of the source dataset across different genders, ethnicities, and facial structures, rather than collapsing into a single generic infant appearance. The skin texture transformation is also consistent across all examples. Each output shows the softer, smoother skin texture characteristics of infant faces, with reduced pore visibility and facial lines compared to the adult inputs. Beyond skin texture, the geometric transformation of facial structure is clearly visible. The synthetic outputs display proportionally larger foreheads relative to face size, smaller and rounder noses, rounder facial shape, and reduced jawline definition, these are all facial characteristics and proportions of infant facial geometry that vanilla style transfer was unable to achieve, as described in Section 3.5. In the output images the blur effect appears only in the background rather than the facial area, suggesting that this method works specifically for faces without generating any artefacts in their image. However, one major drawback of SAM is that it does not fully account for hair transformation, which leads to adult hairstyles being preserved in synthesised images. This is a known constraint of StyleGAN-

based age regression models that primarily work within the facial feature space. Additionally, some outputs appear to represent young children of around 3-5 years old rather than specifically 0-2. This reflects the latent space geometry constraint described in Section 3.5, a limitation that a model designed specifically for infant generation or additional iterations could address in future work. Despite these limitations, the pipeline demonstrates strong scalability as it produces 30,000 synthetic infant images without requiring any real infant photographs, circumventing the ethical and privacy constraints that make real infant facial data difficult to collect at scale, as established in Section 2.3. Overall, the synthetic dataset is a meaningful and innovative contribution. As far as we know, it is the first publicly available synthetic infant facial database created without directly using real images of infants. This database provides a convenient approach for overcoming the problem of the lack of real infant data that hinders previous research in this field.

### 5.2 Image Inpainting Evaluation

#### *5.2.1 Quantitative Results*

The synthetic infant trained inpainting model is evaluated by comparing the inpainted infant output with the output of the pre-trained PUT model from the same infant input. This highlights whether retraining the model on a synthetic infant dataset leads to improvements in its performance for infant inpainting. The retrained PUT model is evaluated on the 2,000 image synthetic infant test set using three metrics: PSNR, SSIM, and LPIPS, each capturing a different aspect of the reconstruction quality. PSNR is calculated to measure pixel-level reconstruction quality between the inpainted output and the original unmasked ground truth image. The retrained PUT achieves a PSNR of 29.86 dB, which is close to the 30 dB threshold commonly associated with high-quality reconstruction in the image inpainting literature [45]. SSIM, which measures structural similarity between the inpainted output and the ground truth by capturing luminance, contrast, and structural information, correlates more closely to human visual perception than PSNR [45]. The retrained PUT achieves an SSIM of 0.9536, indicating strong structural fidelity between the inpainted outputs and the ground truth infant faces. To measure the perceptual similarity between the output and ground truth, where perceptual plausibility matters more than pixel-level accuracy, we compute LPIPS [46]. The retrained PUT achieves an LPIPS of 0.1057, indicating that the inpainted outputs are perceptually plausible reconstructions of the masked facial regions. The pretrained PUT baseline, trained on FFHQ without any domain adaptation to infant faces, was evaluated on the same test set and achieves a higher LPIPS of 0.1164 compared to the retrained 0.1057, while the pretrained model scores marginally higher on PSNR at 30.22 versus 29.86 and SSIM at 0.9581 versus 0.9536, as shown in Table 4. The marginal PSNR and SSIM advantage of the pretrained model is likely because of its significantly larger training set, the original PUT was trained on 70,000 FFHQ images compared to 8,000 synthetic infant images used in this project. The more training data used generally produces better pixel-level reconstruction quality. The LPIPS advantage of the retrained model is more meaningful for this task, as it suggests the domain-adapted model produces outputs that are perceptually more consistent with infant facial appearance despite being trained on considerably less data. Overall, the different models perform similarly, suggesting that the retrained PUT performs comparably to the pretrained baseline while operating in the infant facial domain, which the pretrained model was not designed for.

However, the HINT model which was also retrained using the same 8,000 synthetic images as our retrained PUT model, was also tested to compare our retrained model to another transformer based inpainting model. The retrained HINT model achieves a meaningfully higher PSNR and marginally better LPIPS than the retrained PUT, while the retrained PUT achieves a higher SSIM. HINT having a PSNR 2.01 above our model suggests that its inpainted pixels are numerically closer to the ground truth. Additionally the HINT model beats our model in perceptual similarity as the LPIPS score of the HINT model is 0.0037 lower than our model. This represents HINT outputs being perceptually slightly more similar to the ground truth. The improved PSNR and LPIPS could be because HINT was specifically designed for high fidelity reconstruction, as its mask aware pixel-shuffle downsampling and spatially activated channel attention were explicitly optimised to preserve fine detail and perceptual quality. Even if the HINT architecture was halved due to computational constraints on the Durham NCC cluster, these intentional design choices give it an inherent advantage at pixel-level and perceptual accuracy. These results do not fully reflect the full capacity of HINT's performance as it is not at its full size. PUT's patch-based approach processes images in non-overlapping patches without spatial downsampling, meaning it preserves the global structural relationships across the face more faithfully. This can be seen with PUT reaching a SSIM score 0.0036 higher than HINT. As SSIM captures luminance, contrast, and structural patterns at a global level, PUT is suggested to be more optimised for structural awareness than HINT. For facial reconstruction tasks specifically, structural fidelity is particularly meaningful as clinical predictions must preserve the correct facial geometry, proportions, and spatial relationships of the face rather than simply matching pixel values. PUT's higher SSIM suggests its outputs are more structurally coherent, which for a surgeon viewing a predicted post surgical outcome is arguably more important than marginal pixel-level accuracy. The architectural differences between these two transformer models produce marginal variations in how they handle reconstruction but neither model has a fundamentally different view of the infant domain as they are trained on the same data, giving them a more accurate comparison. Despite these differences, the overall scores across all three metrics are close enough that neither model is definitively superior. This suggests that domain adaptation through retraining on synthetic infant data is effective regardless of which transformer architecture is chosen, and that the synthetic dataset is sufficient to meaningfully adapt both models to the infant facial domain. The idea that a reduced HINT still marginally outperforms PUT on PSNR and LPIPS advocates for HINT's architectural design. If HINT had been trained at full capacity, the difference between model scores might have been more pronounced.

#### 5.2.2 Qualitative Evaluation on Random Masks

In the case of random masking, the retrained PUT produces reliable reconstruction results in all six samples, where the areas reconstructed maintain consistent facial structure and feature placement despite the large and varying shapes of masks. There are some discrepancies in certain samples.

The Chen et al. [14] feasibility model produces the weakest reconstructions, consistent with it being trained on CelebA rather than infant specific data. The cleft feasibility model displays anatomically implausible outputs for infants where duplicated facial features like the lower lip can be seen, incorrectly positioned eyes, and expression inconsistencies not seen in other models. These failures demonstrate the importance of infant-specific domain adaptation, reinforcing the value of the infant-specific domain adaptation and synthetic dataset developed in this project.

The pretrained model exhibits notable issues with eye reconstruction in several examples, for example the leftmost image in Figure 10a shows an incorrectly coloured and disproportionately large pupil in the inpainted eye, and similarly the fourth example shows a flat uncanny eye that lacks the three-dimensional quality of the surrounding face. The second example shows the pretrained model generating adult teeth visible on the lower jaw whilst the ground truth shows no visible teeth, making our retrained model more faithful to the ground truth anatomically. The third example shows the pretrained model creating a slimmer, more defined jawline and cheekbone structure than the ground truth, making the face appear more adult-like. The fourth example also shows a nose shape that is significantly slimmer than the ground truth, an important anatomical detail given the proximity of the nose to the cleft lip region. In contrast, the retrained PUT creates more accurate facial geometry across examples whilst maintaining rounder infant facial proportions, more appropriate nose width, and more naturally shaped eyes consistent with infant anatomy. Minor limitations are the background artefacts as seen in the fifth example, and less defined lip structure in some outputs compared to the pretrained model. The second example in particular shows the pretrained model producing a more defined lower lip. These observations are in line with the quantitative LPIPS advantage of the retrained model, suggesting that domain adaptation to infant features produces outputs that are perceptually more consistent with genuine infant appearance.

The retrained HINT model was able to create visually convincing reconstructions across all six random mask examples, with coherent inpainted regions that are consistent with the surrounding facial context and skin tone. In Figure 10a, example images 1, 4 and 6 for both the retrained HINT and our model produced near-identical results in regards to infant facial structure as well as infant-like appearance. This visual similarity in result quality follows the small quantitative difference seen in Table 4. However, in the third example, HINT is seen to be producing a noticeably stronger reconstruction than PUT, with the curvature and proportions of the head appearing more akin to the ground truth. The retrained PUT however occasionally struggles with predicting the correct head shape as seen in images 2 and 5. Despite HINT's stronger head shape reconstruction, a key architectural distinction is that PUT is a pluralistic inpainting model, meaning that it is capable of generating multiple different plausible reconstructions for the same masked input, unlike HINT which produces a single deterministic output. For cleft lip reconstruction this is a clinically meaningful feature. As there is no single correct post-surgical outcome and these are all predictions, a surgeon would benefit from seeing multiple plausible predictions rather than one deterministic reconstruction as this better represents the range of achievable outcomes and supports more informed surgical planning. PUT's pluralistic capability therefore offers a practical clinical advantage that metrics alone do not capture, and when combined with its stronger structural fidelity as seen by SSIM, this presents PUT as a well-suited model for the uncertainty inherent in surgical outcome prediction.

### 5.3 Qualitative Evaluation on Cleft Lip Masks

The results for the general cleft area mask seen in Figure 10b are particularly encouraging. Across all six examples, the model reconstructs the lip region to produce a natural and continuous mouth area, with the critical clinical detail of no joining between the lip and nose correctly predicted in every case. The first example is particularly strong as the inpainted lip regions appear almost indistinguishable from the ground truth in terms of skin tone, lip shape, and natural curvature. In each output, the reconstructed lip areas match the skin tone and texture of the rest of the face, and the upper lip is present and naturally shaped with a cupid's bow. This shows that the model has learned to predict a plausible infant lip anatomy based on surrounding facial context. The second example produces the least convincing reconstruction of the six, though the output still correctly predicts a continuous lip structure without the cleft deformity. The strong performance under cleft-specific masks is notable given that the model was trained exclusively on randomly generated masks, demonstrating that the domain adaptation to infant facial features generalises beyond the specific mask distribution seen during training. Overall, the inpainting results demonstrate that the retrained PUT model produces convincing reconstructions of synthetic infant faces under both random and cleft-specific mask conditions, with the cleft area results providing strong qualitative evidence that the approach is viable for predicting post-surgical cleft lip outcomes.

#### 5.3.1 Cleft Lip Inference on Cleft Lip Masks

Figure 11 shows the qualitative cleft lip inference results on three infant images using our retrained PUT model, where the model produces anatomically plausible lip reconstructions replacing the masked region with a continuous lip structure free of cleft deformity. All three outputs possess two separate nostrils, which is a significant detail because the cleft lip condition can affect the base of the nose. The third example is particularly compelling because of the well-defined upper lip that blends naturally with the surrounding skin and facial structure. The first example shows slight blurring in the inpainted region, and the second example similarly shows a continuous lip structure but with a less defined upper lip. Both examples reflect

the challenge of generalising from synthetic training data to real photographs with varying size and shape of the cleft region and corresponding mask, which affects how much surrounding facial context and the model can use to predict reconstruction. Despite these variations, the results provide strong qualitative evidence that the pipeline generalises to real clinical images despite being trained entirely on synthetic data with no real infant images used at any stage of training. Quantitative evaluation would require having access to pre and post-surgical ground truth images that can be used for comparison through standard metrics of reconstruction performance (PSNR, SSIM, LPIPS), while distributional comparison (FID to the post-surgical group) could also be measured. Expert surgeon ratings would give a complementary clinical evaluation with the automated metrics. This remains a target direction for future work.

### 5.4 Software Engineering Process

The project utilised an experimental, iterative method rather than the typical software engineering lifecycle approach. The synthetic data generation pipeline (Section 3.5) was iterated through four stages: neural style transfer, landmark warping with style transfer, LoRA-tuned diffusion, and SAM-based age regression - with each stage motivated by weaknesses identified during prior experimentation. Each experiment was conducted separately in an isolated virtual environment on the Durham NCC cluster and included repository clones specifically for each new experiment so candidate approaches could be experimented with and rejected without risking the integrity of other data pipelines. Project management relied on a Gantt chart, experimental log, and meetings log maintained throughout the duration of the project with supervisor meetings every week depending on scheduling. The biggest challenge during this project was the lack of computing power – training larger models took several days per run on NCC, insufficient disk space prevented downloading datasets like IMDB-WIKI to perform additional evaluations and memory limitation required reducing batch sizes from 24 to 4, leading to restricting retraining from the initial goal of 30,000 to 8,000 synthetic images. In retrospect, spending less time on the synthetic data generation phase would have left more time in the inpainting phase to explore architectural extensions beyond retraining alone.

### 5.5 Discussion

The research question of this project was whether a generative image inpainting model trained on synthetic infant data could predict realistic post-surgical outcomes for cleft lip patients, the results of this project provides an encouraging affirmative answer that it can. The synthetic infant dataset generation pipeline successfully produces 30,000 visually coherent infant-like facial images without requiring any real infant images, and the retrained PUT model demonstrates a strong capability for inpainting on synthetic infant faces while being able to generalise to real cleft lip images at inference. The project was subject to computational constraints that limited the synthetic training set to 8,000 images rather than the full 30,000 and required architectural simplification of the HINT model. Larger training sets and longer training times would likely improve performance further. The main limitation of this approach is that quantitative evaluation of the cleft lip inference is not truly possible without paired pre and post-surgical ground truth images. Having access to a clinical dataset would enable more rigorous evaluation and is a target direction for future work. Despite these limitations, this project demonstrates that the combination of synthetic data generation and domain-adapted inpainting represents a viable and ethically motivated approach to cleft lip surgical outcome prediction, contributing a novel synthetic infant dataset and an infant-specific inpainting model that could support future research in this clinically meaningful area.

## 6 Conclusion

This project set out to address a well-documented gap in cleft lip surgical outcome prediction, the absence of publicly available infant-specific data and domain-adapted inpainting models. Through the combination of style-transfer-based methods to generate synthetic infant data and a transformer-based image inpainting model, we demonstrate that it is possible to construct a usable infant-specific reconstruction pipeline entirely from publicly available tools and datasets, without requiring access to real infant photographs or clinical data. The retrained PUT displays a strong inpainting performance with a PSNR of 29.86, an SSIM of 0.9536, and an LPIPS of 0.1057. It performs as well as the pretrained baseline and performs competitively alongside the retrained HINT model, demonstrating that the synthetic dataset effectively adapted inpainting transformer architecture to the infant domain. However, our model was able to be competitive even whilst being trained on significantly less data. Qualitative results based on the use of actual cleft lip pictures revealed that the network is capable of predicting reasonable surgical outcomes without actual infant data.

To the best of our knowledge, this is the first publicly available dataset of synthetic infant faces that are not derived from actual infant images. This is a direct solution to the ethical problems and data scarcity issues that had hampered previous approaches. Together, the pipeline shows a realistic and ethical way of predicting the outcomes of cleft lip surgery that can help surgeons prepare for the procedure and avoid revisions.

The primary limitation is that there were no preoperative and postoperative ground truth images, thus preventing the quantitative evaluation of the cleft lip inference task. A secondary limitation was the computational constraints which limited the synthetic training dataset to 8,000 images and required architectural simplification of the HINT model. However, despite all these limitations, the retrained HINT achieved competitive results suggesting that at full capacity, retraining would yield further improvements. Further work could include utilising clinical datasets officially to quantify the results, expanding the dataset to reach all 30,000 images, and exploring models designed specifically for infant facial generation to improve the accuracy of age regression. The pipeline could also be extended to bilateral cleft lip and cleft palate cases.

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